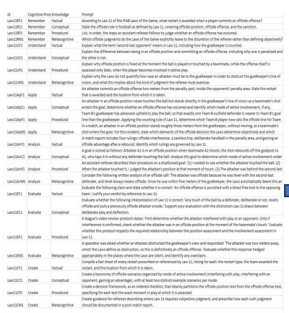
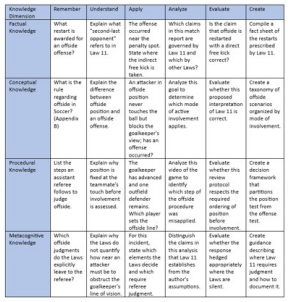
Graphical Abstract

Evaluator Patterns: Creating Standard Contractual Evaluation

Granville Miller

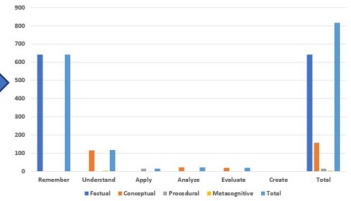


Benchmarks

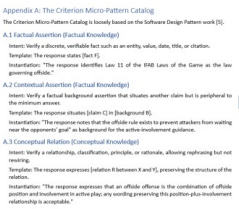


Knowledge Dimension	Remember	Understand	Apply	Analyze	Evaluate	Create
Factual Knowledge	What restart is awarded for an offside offense?	Explain what "second-last opponent" refers to in Law 11.	The offense occurred near the penalty spot. State where the indirect free kick is taken.	Which claims in this match report are governed by Law 11 and which by other Laws?	Is the claim that offside is restarted with a direct free kick correct?	Compile a fact sheet of the restarts prescribed by Law 11.
Conceptual Knowledge	What is the rule regarding offside in Soccer? (Appendix B)	Explain the difference between offside position and an offside offense.	An attacker in offside position never touches the ball but blocks the goalkeeper's view; has an offense occurred?	Analyse this goal to determine which mode of active involvement applies.	Evaluate whether this proposed interpretation of Law 11 is correct.	Create a taxonomy of offside scenarios organised by mode of involvement.
Procedural Knowledge	List the steps an assistant referee follows to judge offside.	Explain why position is fixed at the teammate's touch before involvement is assessed.	The goalkeeper has advanced and one outfield defender remains. Which player sets the offside line?	Analyse this video of the game to identify which step of the offside procedure was misapplied.	Evaluate whether this review protocol respects the required ordering of position before involvement.	Create a decision framework that partitions the position test from the offense test.
Metacognitive Knowledge	Which offside judgments do the Laws explicitly leave to the referee?	Explain why the Laws do not quantify how near an attacker must be to obstruct the goalkeeper's line of vision	For this incident, state which elements the Laws decide and which require referee judgment.	Distinguish the claims in this analysis that Law 11 establishes from the author's assumptions.	Evaluate whether the response hedged appropriately where the Laws are silent.	Create guidance describing where Law 11 requires judgment and how to document it.

Taxonomy Table



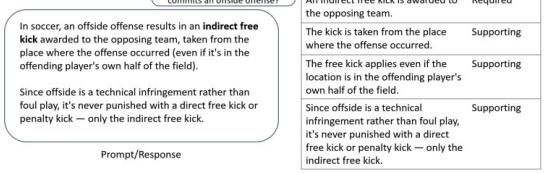
Benchmark Diversity



Evaluator Patterns



Evaluators



Claim	Required/ Supporting
An indirect free kick is awarded to the opposing team.	Required
The kick is taken from the place where the offense occurred.	Supporting
The free kick applies even if the location is in the offending player's own half of the field.	Supporting
Since offside is a technical infringement rather than foul play, it's never punished with a direct free kick or penalty kick — only the indirect free kick.	Supporting

Responses

Highlights

Evaluator Patterns: Creating Standard Contractual Evaluation

Granville Miller

- This paper introduces Evaluator Patterns, a method for standardizing evaluators across prompts, models, and test cycles for Generative AI systems.
- This paper describes explicit, human-auditable evaluators that provide greater traceability than unconstrained LLM-as-a-Judge evaluation.
- This paper presents a mechanism where evaluation and scoring are applied to the whole response rather than individual claims.
- Reference-Guided LLM-as-a-Judge evaluates the presence or absence of these criteria automatically in a single evaluation pass.
- This paper creates evaluators whose changes can be tracked and are versionable like code.
- This paper utilizes Reference-Guided LLM-as-a-Judge combined with evaluators created from Evaluator Patterns to gain greater control over evaluation.
- This paper expands Bloom's Taxonomy Table to categorize and promote diversity among prompts used in testing Generative AI solutions and develop evaluator criteria based on Bloom's Knowledge Dimension.

Evaluator Patterns: Creating Standard Contractual Evaluation

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Abstract

Prompts establish requirements for expected responses in the automated testing of generative AI (GenAI) systems. Models generate the responses, often adding supporting context, explanation, chain-of-thought explanations provided in the response (Wei et al., 2022), or elaboration beyond the minimum answer. Evaluators determine whether a response satisfies those prompt requirements. Because testing begins with prompts collected into benchmarks, prompt construction is a critical test-engineering activity. To that end, the revised Bloom's Taxonomy provides a useful framework for classifying prompts along two separate dimensions, the Cognitive Process and Knowledge Dimensions (Anderson and Krathwohl, 2001; Krathwohl, 2002). These two dimensions provide a mechanism for understanding what is expected from a response and, more importantly, how to evaluate that response.

This paper presents evaluator patterns as contractual specifications connecting prompt engineering, response claims, criterion micro-patterns, and scoring rules using Bloom's Taxonomy. Bloom's Cognitive Process Dimension classifies the operation demanded by a prompt using six categories: Remember, Understand, Apply, Analyze, Evaluate, or Create. Bloom's Knowledge Dimension classifies the kinds of knowledge expected by a prompt and answered by a claim (Song, Kim, and Iyyer, 2024) in the response. The Knowledge Dimension consists of four categories: Factual, Conceptual, Procedural, or Metacognitive. The two dimensions intersect in the revised Taxonomy Table (Krathwohl, 2002). Using these two dimensions, prompts and claims are classified in a manner that allows the predictability and standardization of evaluators.

To create evaluators from evaluator patterns, subject matter expertise, documents, and knowledge graphs created from multiple sources (Miller,

2026a) provide an "a priori" approach to creating criteria. Collected responses, divided into claims (Wanner et al., 2024), provide an "a posteriori" approach to creating criteria. Reference-guided LLM-as-a-Judge (RGLLM-a-a-J) applies the resulting evaluator to score a complete response. The result is a traceable, versioned, and auditable contract for evaluating responses to GenAI benchmarks. Evaluator patterns offer five additional benefits. Evaluation is standardized across prompts, models, and test cycles. The evaluators are human-auditable. Evaluation and scoring are applied to the whole response rather than on individual claims. Scoring is faster than human audit for complex responses. Finally, Bloom's Taxonomy provides a method for achieving prompt diversity within benchmarks.

Keywords:
Evaluator Patterns, Claims, Bloom's Taxonomy, Reference-Guided LLM-as-a-Judge, Benchmark diversity, Benchmark contamination

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1. Introduction

Prompts are the basic unit of interaction in most LLM-based systems, which makes them a natural starting point for testing. In a testing context, a prompt is not merely a user question. A prompt is a structured instrument for eliciting evidence about system behavior. A well-designed prompt can expose failures in retrieval, reasoning within domain boundaries, instruction following, hallucination resistance (Miller, 2026a), output formatting, and synthetic data construction (Chen et al., 2024). A weak prompt may conceal those same defects. Prompt design should therefore be treated as a test-engineering activity rather than an ad hoc writing exercise.

Bloom's Taxonomy (Anderson and Krathwohl, 2001; Krathwohl, 2002) provides insight into both prompts and responses. Using Bloom's Taxonomy, prompts can be mapped onto both the Cognitive Process Dimension (defined in Figure 1) and the Knowledge Dimension (defined in Table 2). Independently, ground truth, in the form of subject matter expertise, documents, knowledge graphs (Miller, 2026a), and audited claims from collected responses can be mapped to Criterion Micro-Patterns within an evaluator corresponding to the prompt. The mapping from ground truth to Criterion Micro-Patterns utilizes Bloom's Taxonomy Knowledge Dimension. As a result, these evaluators become reference guides for scoring responses in automated testing.

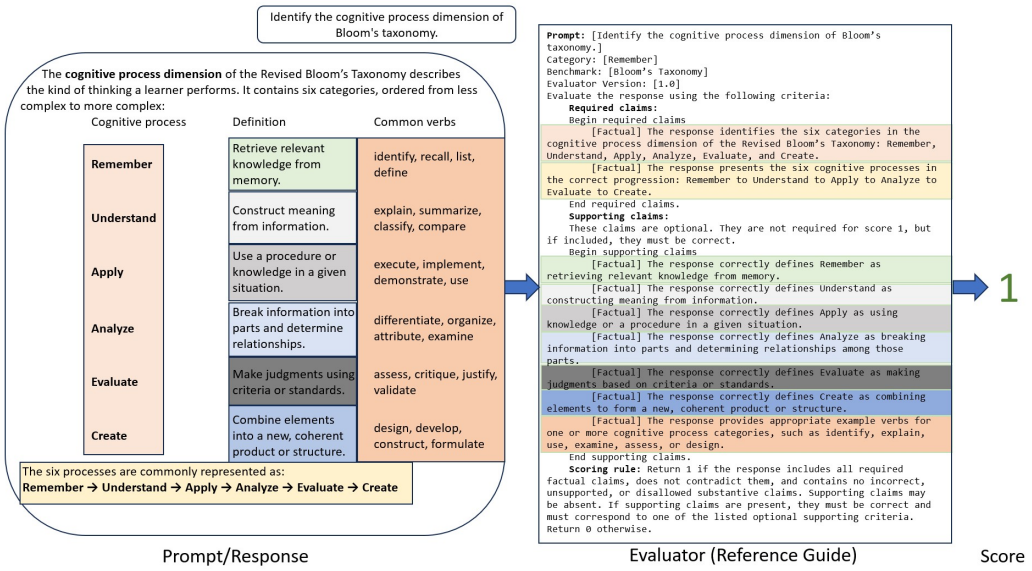


Figure 1: Evaluating a Response Using a RGLLM-a-a-J Evaluator

The product of this effort is evaluator patterns, an abstract contractual specification used to create concrete evaluators for responses from prompts provided to LLM-based systems. Evaluator patterns consist of a scaffold and a set of criterion micro-patterns. Within the scaffold are the overarching rules of how a response must be constructed and one or more scoring rules. The patterns presented in this paper are by no means exhaustive. There are many ways to structure the input and output to generative AI (GenAI) systems which are not covered by these patterns. The evaluator patterns described in this paper focus on prompts structured as information directives and questions. Bloom's taxonomy provides a framework for classifying these sorts of prompts and their resulting claims. However, similar evaluator patterns could be built for other taxonomies classifying interactions such as conversational chat.

Evaluator patterns provide five benefits to enterprise GenAI testing. First, evaluators constructed from evaluator patterns provide standardization, auditability, and versioning. Every evaluator is derived from a common contractual specification, so criterion micro-patterns and scoring rules are constructed and applied uniformly across prompts, models, and test cycles. Second, the resulting evaluators are human-auditable. Since the criteria and

scoring rules are explicit, traceable, and versioned, a human reviewer can inspect exactly which criteria a response satisfied or failed and why it received its score. Third, the approach applies the evaluator to the whole response. Evaluating the whole response is in contrast to approaches requiring a response to be divided into individual claims at test time so that individual claims can be verified (Min et al., 2023). The RGLLM-a-a-J evaluates the complete response against all applicable evaluator criteria in a single pass, without requiring claim decomposition at test time. This score includes supporting context, chain-of-thought explanations provided in the response, and elaboration beyond the minimum answer. Fourth, evaluation is faster than human audit for complex responses. Once an evaluator is constructed, scoring long, multi-claim responses is automated. Review effort does not grow significantly with response length or complexity as it does with manual audits. Finally, evaluator patterns provide a method for prompt diversity. Mapping prompts onto the Cognitive Process and Knowledge Dimensions of Bloom's Taxonomy exposes gaps in benchmark coverage and guides the construction of prompts that span the full Taxonomy Table, promoting the AI equivalent of both retention and transfer (Mayer and Wittrock, 1996).

2. Prior Work

Bloom's Taxonomy was developed in the 1950s by a team of researchers led by Benjamin Bloom. During that period, as much as 90% of instructional time in education was spent at the "Knowledge" (now called "Remembering") level (Bloom, 1994). Since the taxonomy was aimed at classifying the goals of the education process, this meant most education was focused on knowledge retention and not on transfer (Mayer, 2002). Retention is the ability to remember material at a later time. Transfer is the ability to use what was learned to solve new problems (Mayer and Wittrock, 1996). Bloom's Taxonomy identified a two-dimensional table for classifying educational objectives that emphasized both retention and transfer.

> "The group believed that some common framework used by all college and university examiners could do much to promote the exchange of test materials and ideas for testing. They also believed that such a framework could be useful in stimulating research on examinations and on the relation between examinations and education. After considerable discussion, there was agreement that

> the framework might best be obtained through a system of classifying goals of the education process using educational objectives."
> – B. Bloom (Bloom, 1994).

Bloom's Taxonomy is commonly applied to educational objectives. The taxonomy can also be applied to information directives and questions that can be characterized in verb-noun form, commonly found in most benchmarks. Bloom's Taxonomy Cognitive Process and Knowledge Dimensions can serve as a reminder that benchmarks should be balanced to promote the AI equivalent of retention and transfer. Figure 2 shows the distribution of prompts for the popular open-source benchmark, TruthfulQA (Lin, Hilton, and Evans, 2025). Of the 817 questions in the benchmark, 643 prompts fall in the Remember x Factual Knowledge Dimension (see Table 1). The reason for commonly using these knowledge retrieval prompts is that they are often the easiest to evaluate. The thinking may be that "if the knowledge retrieval is there, the knowledge transfer will follow". In addition, responses to a Remember x Factual prompt are often the shortest and easiest for a human to read and evaluate.

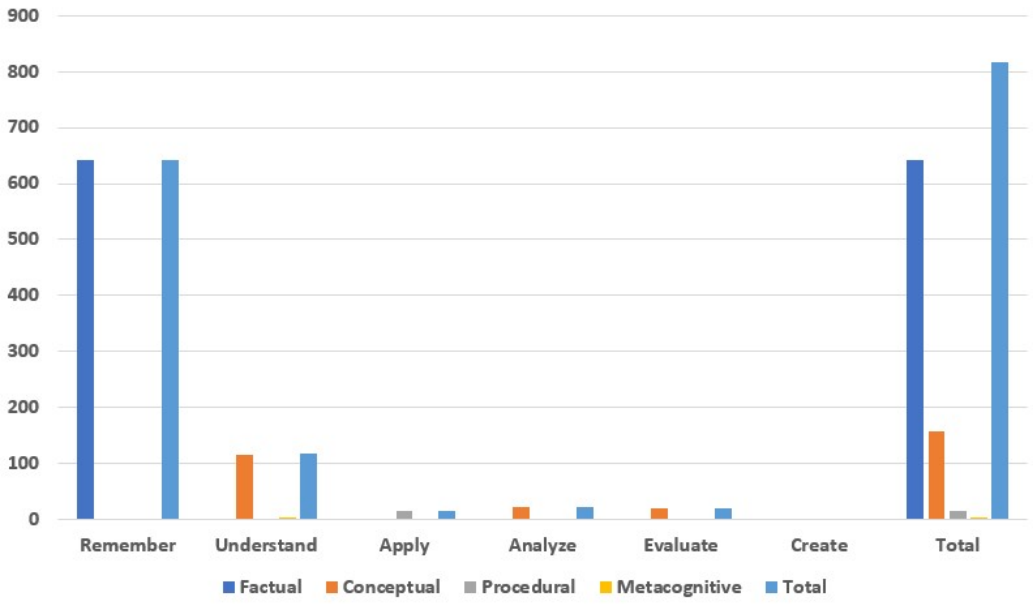


Figure 2: TruthfulQA questions (Lin, Hilton, and Evans, 2025) by Bloom's Taxonomy Dimensions (see Table 1)

Once balanced benchmarks are created, the responses to the prompts in a benchmark must be evaluated as well. LLM-as-a-Judge (LLM-a-a-J) is a common method of evaluating responses that provides a scalable alternative to human review. Zheng et al. introduced this widely used evaluation paradigm with MT-Bench and Chatbot Arena and reported strong agreement between capable judge models and human preferences (Zheng et al., 2023). The same work documented systematic weaknesses in LLM-a-a-Judge, including position bias, verbosity bias, self-enhancement bias, and limitations in grading some reasoning tasks (Zheng et al., 2023).

These weaknesses are magnified in enterprise settings. Straight LLM-as-a-Judge asks the judge to determine correctness primarily from its parametric knowledge. That knowledge is largely fixed at training time and may not contain the proprietary, domain-specific, or frequently changing information against which an enterprise response must be measured. The responses may contain proprietary company policies used in help desk systems or corporate action memoranda used in decision support which may be unknown to the judge. Even when both the system under test and the judge have the same authoritative enterprise information, the holes in that information that produce an incorrect response (Miller, 2026a) can also produce an incorrect verdict. Zheng et al. showed that supplying the judge with a reference answer substantially reduces grading failures on questions with objective solutions (Zheng et al., 2023). For these reasons, this paper builds on reference-guided LLM-as-a-Judge rather than straight LLM-as-a-Judge.

3. Bloom's Taxonomy for Prompts

Bloom's revised taxonomy arranges its two dimensions into the Taxonomy Table (Krathwohl, 2002). The table is a grid in which the four categories of the Knowledge Dimension form the columns and the six categories of the Cognitive Process Dimension form the rows (Anderson and Krathwohl, 2001; Krathwohl, 2002) (see Table 1 and Appendix C). A prompt is placed in the table by parsing its verb and its noun. The verb names the cognitive operation and selects the row, while the noun names the kind of knowledge operated on and selects the column (Krathwohl, 2002). Questions or directive prompts of the verb-noun form found in most benchmarks follow the sentence format that may be parsed to provide this row and column. For example, "What is the rule regarding offside in Football?", the prompt evaluated by the evaluator in Appendix B, asks the responder to recall (Remember) the offside

rule. Although the offside rule contains interrelated concepts, this prompt asks the model to recall the rule as stated in the reference material. For purposes of this benchmark, the prompt is therefore classified as Remember × Factual (see Table 1).

The cell a prompt occupies does not guarantee that the model will behave as expected when prompted. The Knowledge Dimension characterizes the knowledge demanded by the prompt rather than the internal mechanism by which the model produces its response. Consider the prompt in the Apply × Procedural cell (see Appendix C): "Team B's goalkeeper has advanced upfield to play the ball, so that exactly one Team B outfield defender is nearer to Team B's goal line than the goalkeeper. Applying the counting rule of Law 11, determine which Team B player now sets the offside line for Team A's attackers." A model that has stored the fact that the second-last opponent may be an outfield player answers by retrieval, and for that model the prompt is not executing the transfer expected by the tester. A model lacking that fact must reason it out, applying the counting rule of Law 11 (IFAB, 2026), the second-last opponent with the goalkeeper counted among the defenders, to derive the answer as the output of a Procedural Transformation. Benchmark contamination occurs when benchmark items or closely related material are present in a model's training data, allowing benchmark performance to reflect prior exposure rather than solely the capability the benchmark is intended to measure (Sainz et al., 2023; Brown et al., 2020). In these cases, the full impact of even the toughest prompts is not felt by the system under test.

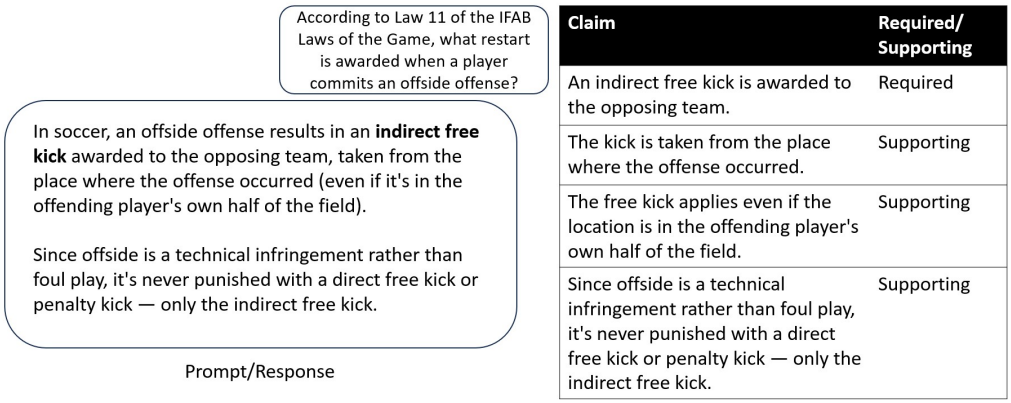


Figure 3: A "Remember" Prompt looking for Factual Knowledge

Knowledge Dimension	Factual	Conceptual	Procedural	Metacognitive
Remember	According to Law 11 of the IFAB Laws of the Game, what restart is awarded when a player commits an offside offense?	State the offside rule in football as defined by Law 11, covering offside position, offside offense, and the sanction.	List, in order, the steps an assistant referee follows to judge whether an offside offense has occurred.	Which offside judgments do the Laws of the Game explicitly leave to the discretion of the referee rather than defining objectively?
Understand	Explain what the term "second-last opponent" means in Law 11, including how the goalkeeper is counted.	Explain the difference between being in an offside position and committing an offside offense, including why one is penalized and the other is not.	Explain why offside position is fixed at the moment the ball is played or touched by a teammate, while the offense itself is assessed only later, when the player becomes involved in active play.	Explain why the Laws do not quantify how near an attacker must be to the goalkeeper in order to obstruct the goalkeeper's line of vision, and what this implies about the kind of judgment the referee must exercise.

Table 1: The Taxonomy Table for Remember and Understand with Prompts

4. The Enterprise Response Evaluator Scaffold

Creating an evaluator begins with the recognition that even simple factual prompts rarely produce simple factual responses. A prompt such as "According to Law 11 of the IFAB Laws of the Game, what restart is awarded when a player commits an offside offense?" appears to require the validation of only one required claim: An indirect free kick is awarded to the opposing team (see Figure 3). However, the response may also include optional supporting claims, such as the claim that the free kick applies even on the offending player's side of the field or that it is never a direct kick because it is a technical infringement. These additional claims may improve the usefulness of the response, but they also increase the evaluation burden because each additional factual assertion can be correct, unsupported, misleading, or false. An evaluator therefore cannot merely determine whether the required answer appears in the response. The evaluator must also determine whether any optional claims included by the model are accurate and grounded.

Figure 3 shows that both required and supporting claims must be accounted for. As a result, all evaluators in this paper use one scaffold, un-

changed regardless of claim types. Its three sections assign polarity to criteria, and its scoring rule is uniform:

> Prompt: [Prompt]
>
> Category: [Bloom's Taxonomy]
>
> Benchmark: [Benchmark Name]
>
> Evaluator Version: [Version]
>
> Evaluate the response using the following criteria:
>
> Required claims:
>
> > Begin required claims
> >
> > [C1] [criteria pattern] required criteria 1
> >
> > End required claims.
>
> Supporting claims:
>
> These claims are optional. They are not required for score 1, but if included, they must be correct.
>
> > Begin supporting claims
> >
> > [S1] [criteria pattern] supporting criteria 1
> >
> > End supporting claims.
>
> Disallowed claims:
>
> The response must not include any of the following.
>
> > Begin disallowed claims
> >
> > [D1] [criteria pattern] disallowed criteria 1
> >
> > End disallowed claims.
>
> Scoring rule: Return 1 if the response satisfies all required criteria, does not contradict them, and contains no incorrect, unsupported, or disallowed substantive claims. Supporting claims may be absent. If supporting claims are present, they must be correct and must correspond to one of the listed optional supporting criteria. Return 0 otherwise.

The scaffold encodes three design decisions that hold for every claim type. First, polarity is positional. The same criterion content behaves as a gate in the required section, a conditional check in the supporting claim section, and a prohibition in the disallowed section. Second, the supporting section solves the permissiveness/strictness dilemma. The evaluator will not pass a response that adds false context nor fail a response that omits optional context. Third, the scoring rule operationalizes:

> Enterprise Response Conformance Rule #1: If any substantive claim in the response is materially false, unsupported, or disallowed, the response fails the evaluator.

This rule closes the optional claim space. A supporting claim that is present must not only be correct but must correspond to a listed criterion. Novel substantive embellishments are rejected by default rather than passed by omission. A score of zero invokes a human audit, which provides final arbitration and may result in a change to the evaluator or a bug indicating a failure of the system. Everything not established by the scaffold, including what to check, against which references, and in what format, is delegated to the criteria. Although criteria may correspond to individual claims, the evaluator, including its scaffold, criteria, and scoring rule, is applied to the response as a whole.

Note that not every GenAI application requires this strict binary conformance rule. Research assistance, conversation, brainstorming, or fiction may legitimately tolerate partial correctness, uncertainty, or creative variation. Most enterprise systems which provide information to employees and decision makers expect some level of authoritativeness (Miller, 2026a). Evaluator patterns therefore include both criterion structures and scoring rules appropriate to the use case (Armour and Miller, 2001). The binary rule in the provided scaffold is intended for enterprise information systems in which a materially incorrect substantive claim causes the response to fail conformance.

4.1. Applying the Scaffold

To understand the need for the scaffold, a single prompt is run against the same system twice. Each response contains a different set of optional supporting claims. The prompt shown in Figure 4 is classified as Remember $\times$ Factual and so the response should be straightforward. However, the rule itself contains conceptual knowledge which will increase the complexity of the evaluator (see Appendix B). Human auditing of the two responses in Figure 4 is difficult because the long, information-dense responses contain different sets of supporting information.

$$\text{Response } 1 = C_R + \{S_1 + S_2 + S_3 + S_6 + S_7\}$$
$$\text{Response } 2 = C_R + \{S_4 + S_6\}$$

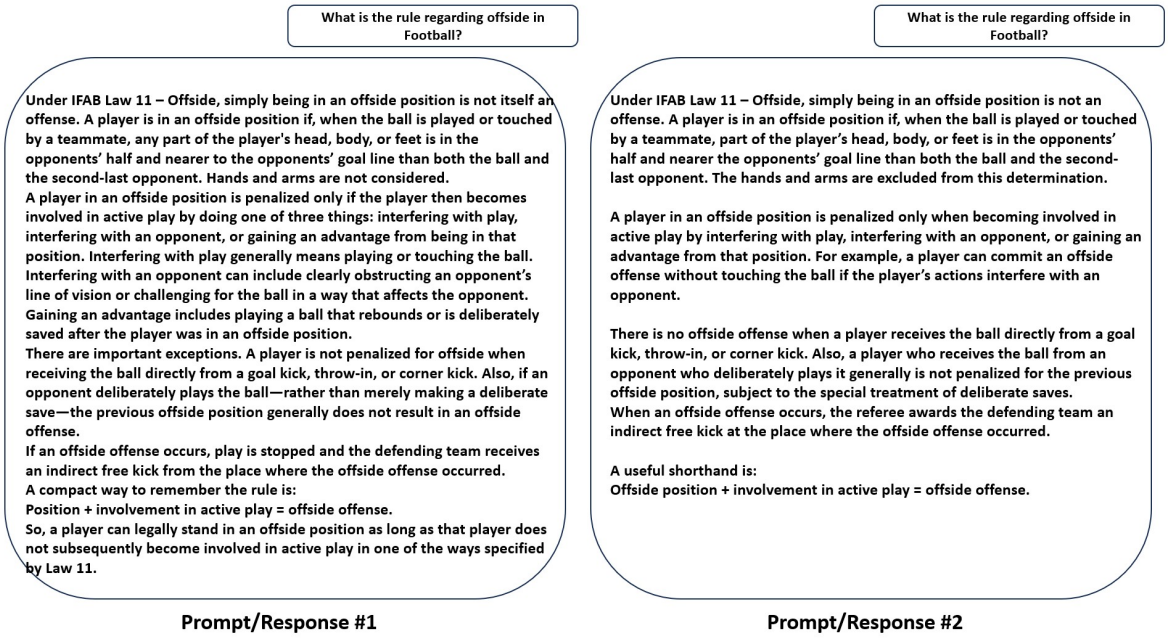


Figure 4: Two different responses from the same prompt and the same system

where C_R represents the required claims, and S_i represents the individual permissible supporting claims specified in the evaluator in Appendix B. RGLLM-as-a-Judge can determine the presence or absence of all of these claims much faster than a human audit. The results of applying the evaluator in Appendix B to Response 1 are shown in Figure 5 (required claims) and Figure 6 (supporting claims).

5. The Criterion Micro-Pattern Catalog

The Criterion Micro-Pattern Catalog in Appendix A describes thirteen Criterion Micro-Patterns. Criterion Micro-Patterns are reusable building blocks from which the criteria in concrete evaluators can be assembled. A criterion is a contractual clause in the evaluator that binds an expected or observed claim to ground truth and renders it testable. The catalog separates the micro-patterns into three functions: claim content, cross-cutting, and response/task conformance (see Table 2). Claim content patterns evaluate the semantic content of claims and are organized by the Knowledge Dimension of Bloom's Taxonomy. The Knowledge Dimension constrains the family

Required Claims

Claim	Status
Identifies IFAB Law 11 – Offside	✅ Pass — "Under IFAB Law 11 – Offside"
Definitional (temporal anchor at teammate's touch; head/body/feet; opponents' half; nearer than ball and second-last opponent; conditions conjoint)	✅ Pass — "when the ball is played or touched by a teammate, any part of the player's head, body, or feet is in the opponents' half **and** nearer to the opponents' goal line than both the ball and the second-last opponent." All components present, judgment correctly anchored to the teammate's touch, conditions stated conjunctively. The halfway-line exclusion is no longer required by this criterion.
Hands and arms excluded	✅ Pass — "Hands and arms are not considered"
Offside position not itself an offense	✅ Pass — first sentence
Offense only via active involvement (three modes named)	✅ Pass — interfering with play, interfering with an opponent, gaining an advantage
Procedural Guard: goal kick / throw-in / corner kick exemption	✅ Pass — stated explicitly as an exception
Procedural Guard: deliberate play generally not penalized, deliberate-save exception preserved	✅ Pass — "if an opponent deliberately plays the ball—rather than merely making a deliberate save—the previous offside position generally does not result in an offside offense"
Indirect free kick to the defending team	✅ Pass
Kick taken from the place of the offense	✅ Pass — "from the place where the offside offense occurred"
Retrieval Presence / Cognitive Operation Conformance	✅ Pass — presents the governing rule itself, not tactics or opinion

Figure 5: Debugging and Scoring the Required Claims of Response #1

of claim content patterns that may be appropriate for the given knowledge type. The Knowledge Dimension does not uniquely determine a single criterion pattern but rather provides multiple criterion micro-patterns (see Table 2). Examples of claims structured by criterion micro-patterns can be found in the evaluator in Appendix B.

Claim content criterion micro-patterns are created from the human research used to create the prompt whose response is being evaluated (a priori). These criteria can also be created from validated claims in the resulting responses (a posteriori). A claim is a unit of substantive content in a response (Min et al., 2023). Like prompts, claims can be classified by Knowledge Dimensions. A claim's Knowledge Dimension describes what kind of knowledge it represents. The Knowledge Dimension constrains the family of claim content micro-patterns appropriate for that criterion, but judging the correctness of a single claim may require more than one pattern. This classification is descriptive and does not determine correctness of the claim. Once a claim

Claim	Status
Interfering with play generally means playing or touching the ball	✅ Present, correct — "Interfering with play generally means playing or touching the ball"
Interfering with an opponent can include obstructing line of vision or challenging in a way that affects the opponent	✅ Present, correct — "clearly obstructing an opponent's line of vision or challenging for the ball in a way that affects the opponent"
Gaining an advantage includes playing a ball that rebounds or is deliberately saved	✅ Present, correct — "playing a ball that rebounds or is deliberately saved after the player was in an offside position"
A player may commit an offside offense without touching the ball via interfering with an opponent	➖ Absent — permitted; no effect on score
Halfway line excluded from the opponents' half	➖ Absent — permitted; no effect on score
Shorthand (position + involvement = offense), presented as a summary not a substitute	✅ Present, correct — offered as "a compact way to remember the rule" after the full rule is stated
A player may legally remain in an offside position absent involvement in active play	✅ Present, correct — "a player can legally stand in an offside position as long as..."
Citation to IFAB Law 11 or equivalent official IFAB material	➖ Absent — permitted; no effect on score

Figure 6: Debugging and Scoring the Supporting Claims of Response #1

is classified by Knowledge Dimension, one or more criterion micro-patterns can be used to create concrete criteria from the claim.

Cross-cutting claim patterns evaluate properties that can qualify claims regardless of the Knowledge Dimension. Cross-cutting claim patterns add properties such as deontic force (whether something is required, prohibited, permitted, optional, or recommended) or citation support. Deontic force and correct citations are just as important to the correctness of a response as the required or supporting claims.

Response/task-conformance patterns operate at a different level from claim content patterns. These patterns apply to the entire response. Cognitive Operation Conformance tests whether the response performs the operation demanded by the prompt and, where applicable, whether the system calls the tools necessary to perform that operation. Ensuring proper tool calling is important for testing LLM systems architectures that employ Native Tool Calling (NTC) for example (Schick et al., 2023). Artifact Structure

Knowledge Dimension	Definition	Criterion Micro-Patterns
Factual Knowledge	The basic, discrete elements that AI must know to be acquainted with a discipline or solve problems. Factual Knowledge is the "bits of information" level, isolated facts that are meaningful on their own without requiring understanding of a larger structure.	Factual Assertion Contextual Assertion
Conceptual Knowledge	Knowledge of the interrelationships among the basic elements within a larger structure that enable them to function together. Where factual knowledge is about isolated bits, conceptual knowledge is about how those bits are organized, the frameworks, categories, and structures that give a domain its shape.	Conceptual Relation Definitional
Procedural Knowledge	Knowledge of how to do something or methods of inquiry, and the criteria for using skills, algorithms, techniques, and methods. Where factual and conceptual knowledge are declarative (knowing that), procedural knowledge is knowing how. Procedural knowledge takes the form of a sequence of steps to be followed.	Procedural Step Presence Procedural Ordering Procedural Transformation Procedural Guard
Metacognitive Knowledge	Knowledge of cognition in general, together with awareness and knowledge of one's own cognition. Metacognitive Knowledge is the reflexive dimension, knowledge about knowledge and knowing, rather than knowledge of the domain's facts, structures, or methods.	Epistemic Status Consistency
Cross-Cutting Claim Properties	Properties that qualify claims regardless of their knowledge dimension.	Deontic Status Consistency Citation Support
Response/ Task Conformance	Criteria that evaluate required cognitive operations or response form rather than the truth of an individual claim.	Cognitive Operation Conformance Artifact Structure

Table 2: Criterion Micro-Pattern Families

tests required response form independent of the truth of individual claims. For example, a prompt may call for a checklist, report, or evaluation of the given subject.

Figure 7 shows the increasing complexity of the criteria necessary to test the responses to the prompts in Table 1 and Appendix C. The figure shows the average number of criteria of the types shown in Appendix A necessary to evaluate their responses. The two-dimensional Bloom classifications shown in Figure 7 correspond to the cells in Table 1 and Appendix C. The averages were determined by running each prompt in Appendix C ten times on GPT-5.6 Sol, Anthropic Fable 5, and Google Gemini 3.7 Flash. The benchmark and results are available on GitHub (Miller, 2026b).

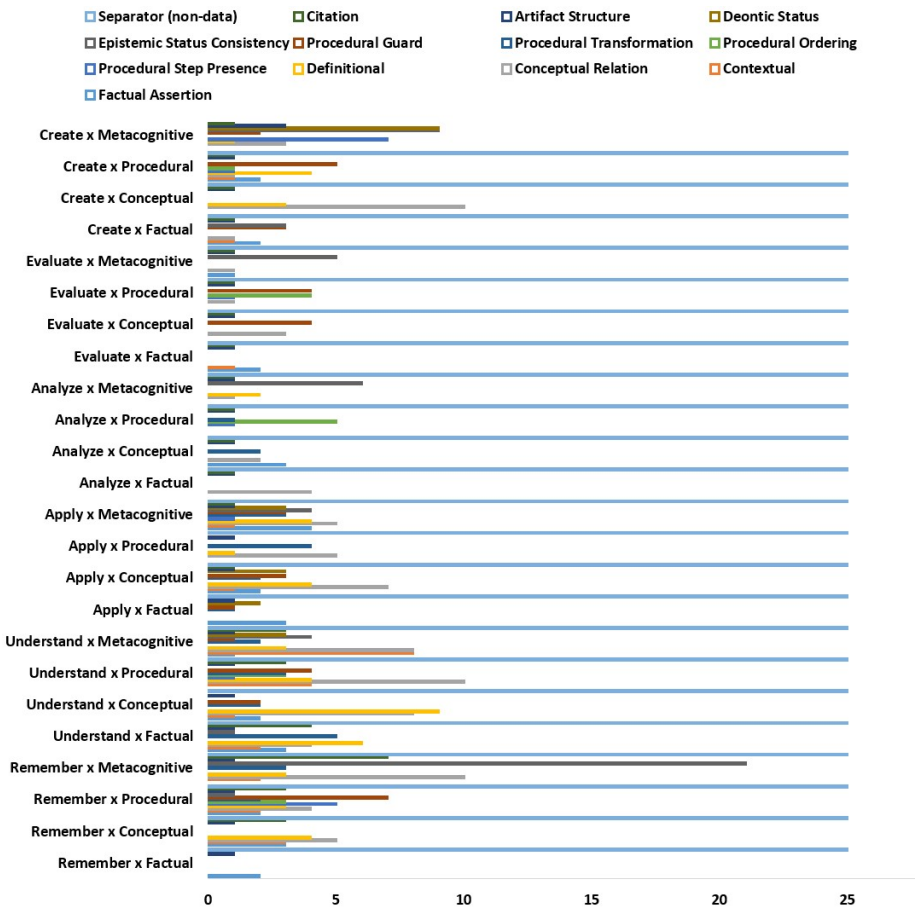


Figure 7: Average Micro-Criteria Types for Claims to Responses for the Prompts in Appendix C

For the Law 11 prompt set examined here, Figure 7 suggests that, as prompts move from Factual toward Metacognitive Knowledge, the number and diversity of criteria used to evaluate their responses generally increase. Although each prompt and response is different, the observed increase in evaluation complexity intuitively follows patterns familiar from the construction of tests for people.

5.1. Creating Criteria

An evaluator is built with the requirements for a given prompt. This concept is important as the prompt should always be the reference point for

the validity of evaluators, claims, and responses. While evaluators evaluate entire responses, they are built and debugged using claims within the responses. There are two methods of creating these evaluators: an "a priori" approach in which evaluators are created before running the prompts and an "a posteriori" approach in which the evaluator is augmented via response data. LLMs can be used to create an evaluator using the rules from this paper which accelerates the initial evaluator-creation process. Regardless of the approach employed, it is critical that all criteria are audited and validated before using them for evaluation.

In the "a priori" approach, criteria are derived from subject matter expertise, text in a document, or related paths in a knowledge graph. When the necessary knowledge comes from a single source, such as a rule book, deriving criteria can be straightforward. When it comes from multiple sources, such as laws or government policies, knowledge graphs can reconcile those sources into a coherent claim space. Knowledge graphs make conflicts, overlaps, and dependencies among provisions explicit so that each derived criterion rests on a single, versioned representation of ground truth rather than on an ad hoc reading of several texts. Constructing knowledge graphs incrementally from knowledge-graph fragments provides one approach to consolidating knowledge from different sources (Miller, 2026a).

A posteriori criteria are harvested from responses after the system is exercised and responses are provided. Harvesting claims follows the same process as debugging evaluators shown in Figures 5 and 6. If a new claim is identified, that claim is verified by the tester using the appropriate reference information. Verified claims that are correct but not covered are the productive case. When a new claim is found, the scoring rule in the evaluator will fail the evaluation, resulting in a human audit. If the claim is verified, the Criterion Micro-Pattern Catalog is used to turn it into a concrete criterion, and the resulting criterion is added to the required or supporting section of the scaffold. If the claim is a hallucination, a bug is created for the system under test.

If the claim is deemed incorrect by the evaluator, the evaluator will also fail the evaluation. If the claim is found to be valid, the corresponding criteria may be wrong. Existing criteria may require some rework to accommodate accurate claims with a broader scope. If the claim is a hallucination, a bug is created for the system under test. In addition, a disallowed criterion can be added.

Finally, required claims may be missing, causing the evaluator to fail.

In these cases, the corresponding required criterion should be examined to determine whether the missing claim is actually required. Use the prompt as a guide and ask, "Is this missing claim really necessary for a response?" If so, write the defect. Missing required information could be a data curation problem. If the criteria is not required, the criteria may warrant a move from the required to the supporting section of the scaffold.

This algorithm guides the evaluator process toward a steady state. The a posteriori loop is the maintenance mechanism that keeps the evaluation contract honest over time. Rapidly changing data such as that used in web grounding may not allow your evaluators to converge. However, enterprises generally have invariant information such as corporate policies which need to remain constant in the face of changing data. The key to good prompt engineering is understanding the needs of the users of the system and designing benchmarks accordingly.

6. Conclusion

This paper introduced Evaluator Patterns as a method for transforming GenAI evaluation from an ad hoc judgment activity into a standardized, contractual testing process. By connecting prompts, response claims, Criterion Micro-Patterns, and explicit scoring rules, the approach makes evaluation traceable, versionable, and human-auditable across prompts, models, and test cycles. The use of Reference-Guided LLM-as-a-Judge provides a scalable mechanism for applying these contracts to complete responses, including required claims, optional supporting claims, and potentially incorrect or disallowed claims. Rather than relying on the judge model's parametric knowledge alone, evaluator criteria can be established "a priori" from subject-matter expertise, authoritative documents, and knowledge graphs and augmented "a posteriori" through validated claims observed in model responses. In this way, evaluator patterns provide both the rigor needed for authoritative enterprise systems and a reusable framework for maintaining evaluation criteria as systems, benchmarks, and knowledge sources evolve.

The evaluator patterns described in this paper are designed to be applied to prompts in the form of questions or information directives. For convergence, they require some level of stability in the data contained in the GenAI system under test. Most enterprise GenAI systems possess this quality. Further research is necessary to understand the types of patterns necessary to test rapidly changing information such as that brought in from

the web. Evaluating time-sensitive prompts, such as those requesting a football player's current scoring statistics, is another area for future research.

A central contribution of this approach is the application of Bloom's revised Taxonomy to GenAI testing. The Cognitive Process Dimension provides a systematic way to characterize what a prompt requires a model to do, Remember, Understand, Apply, Analyze, Evaluate, or Create, while the Knowledge Dimension distinguishes the Factual, Conceptual, Procedural, and Metacognitive knowledge represented in the resulting claims. This framework makes it possible not only to construct more predictable evaluators but also to identify gaps in benchmark design. Existing benchmarks can disproportionately emphasize knowledge retrieval. For example, the paper shows that 643 of TruthfulQA's 817 questions fall into the Remember $\times$ Factual category. Evaluating GenAI solely on such prompts risks testing retention while assuming that transfer will follow. A more complete benchmark should exercise both retention and transfer by deliberately sampling across the Taxonomy Table, thereby requiring systems not merely to retrieve knowledge but to explain relationships, execute procedures, analyze structures, make judgments, and create new artifacts.

Perhaps most importantly, the results suggest that knowledge exhibits similar evaluative patterns in people and GenAI systems. Bloom's Taxonomy was developed to distinguish levels and types of human learning, yet the same structure provides a useful account of the increasing demands placed on generative models. As prompts move from Factual toward Conceptual, Procedural, and Metacognitive knowledge, the number and diversity of criteria needed to evaluate their responses increase. As prompts progress from Remember toward Create, the number of necessary criteria also tends to increase. The paper observes that this progression follows patterns familiar from the construction of tests for people. This does not imply that human cognition and GenAI operate through the same underlying mechanisms. Rather, it suggests that the structure of knowledge and the observable demands of demonstrating that knowledge exhibit comparable patterns. That correspondence is practically significant. Frameworks developed to distinguish human retention, understanding, transfer, and higher-order performance can provide a principled foundation for constructing benchmarks and evaluators that ask the same fundamental question of GenAI systems. These questions emphasize not simply what does the system know, but what can it reliably do with what it knows?

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Appendix A. The Criterion Micro-Pattern Catalog

The Criterion Micro-Pattern Catalog is loosely based on the Software Design Pattern work (Gamma et al., 1995).

Appendix A.1. Factual Assertion (Factual Knowledge)

Intent: Verify a discrete, verifiable fact such as an entity, value, date, title, or citation.

Template: The response states [fact F].

Instantiation: "The response identifies Law 11 of the IFAB Laws of the Game as the law governing offside."

Appendix A.2. Contextual Assertion (Factual Knowledge)

Intent: Verify a factual background assertion that situates another claim but is peripheral to the minimum answer.

Template: The response situates [claim C] in [background B].

Instantiation: "The response notes that the offside rule exists to prevent attackers from waiting near the opponents' goal" as background for the active-involvement guidance.

Appendix A.3. Conceptual Relation (Conceptual Knowledge)

Intent: Verify a relationship, classification, principle, or rationale, allowing rephrasing but not rewiring.

Template: The response expresses [relation R between X and Y], preserving the structure of the relation.

Instantiation: "The response expresses that an offside offense is the combination of offside position and involvement in active play; any wording preserving this position-plus-involvement relationship is acceptable."

Appendix A.4. Definitional (Conceptual Knowledge)

Intent: Verify a definition or the necessary and sufficient conditions attached to a term.

Template: The response defines [term T] by [conditions].

Instantiation: "The response defines an offside position as being nearer to the opponents' goal line than both the ball and the second-last opponent, while in the opponents' half, with the hands and arms of all players excluded from the judgment."

Appendix A.5. Procedural Step Presence (Procedural Knowledge)

Intent: Verify that a mandatory step of a reference procedure appears in the response.

Template: The response includes [step S] of the reference procedure.

Instantiation: "The response includes a step in which the assistant referee fixes the moment the ball is last played or touched by a teammate before judging position."

Appendix A.6. Procedural Ordering (Procedural Knowledge)

Intent: Verify that steps respect the reference procedure's precedence constraints.

Template: The response orders [step S_i] before [step S_j] where the reference requires it.

Instantiation: "The response places the assessment of the attacker's position at the moment of the teammate's touch before the assessment of involvement in active play, since position is frozen at the touch while involvement may develop afterward."

Appendix A.7. Procedural Transformation (Procedural Knowledge)

Intent: Verify that a procedural transformation uses the correct operation and inputs and produces the correct intermediate or final result.

Template: The response derives [output O] from [inputs I] via [operation Op].

Instantiation: "The response correctly determines the offside line as the second-last opponent, counting the goalkeeper among the defenders, so that with a goalkeeper and one outfield defender back, the outfield defender sets the line."

Appendix A.8. Procedural Guard (Procedural Knowledge)

Intent: Verify that the applicability conditions of a step or procedure are stated and respected.

Template: The response conditions [step S] on [guard G].

Instantiation: "The response conditions the offside judgment on the ball not having been received directly from a goal kick, corner kick, or throw-in."

Appendix A.9. Epistemic Status Consistency (Metacognitive Knowledge)

Intent: Verify that asserted certainty and scope match the reference's epistemic status, including acknowledging what the available information does not establish. In this evaluator framework, Metacognitive Knowledge is operationalized as epistemic awareness or knowledge about what is known, uncertain, assumption-dependent, or unsupported.

Template: The response asserts [claim C] with scope and confidence matching [annotation A], and where the necessary information is unavailable, the response states that limitation rather than supplying an unsupported answer.

Instantiation: "Where the Laws do not quantify the asked case (e.g., how near an attacker must be to obstruct a goalkeeper's line of vision), the response states that the Laws leave this to the referee's judgment rather than asserting a threshold."

Appendix A.10. Deontic Status Consistency (Cross-Cutting Claim Property)

Intent: Verify that the binding force of a claim, mandatory, advisory, permitted, or prohibited, matches that of the reference information.

Template: The response presents [claim C] as [modality M].

Instantiation: "The response presents the disallowing of a goal scored directly from an offside offense as mandatory under Law 11," paired with a Factual Assertion criterion on the same claim.

Appendix A.11. Citation Support (Cross-Cutting Claim Property)

Intent: Verify that required citations are present and that cited material supports the corresponding response claims. Citations can be paired with individual claims or the entire response.

Template: The response cites [source S], and [source S] supports [claim C | response R].

Instantiation: "The response cites the IFAB Laws of the Game, and the cited Law 11 supports the response's claim about offside."

Appendix A.12. Artifact Structure (Response/Task Conformance)

Intent: Verify that a created deliverable has the response structure the prompt demands, independent of any individual claim's truth.

Template: The response contains [structural element E] satisfying [property P].

Instantiation: "The response is organized as a decision framework and contains an explicit partition of the position test from the offense test."

Appendix A.13. Cognitive Operation Conformance (Response/Task Conformance)

Intent: Verify that the response performs the cognitive operation required by a prompt requirement, independently of whether its individual claims are factually correct.

Template: The response [retrieves/explains/applies/analyzes/evaluates/creates] [target content or artifact] as required by the prompt.

Instantiation: "For an Evaluate requirement, the response makes an explicit judgment about whether the proposed interpretation of Law 11 is correct and justifies that judgment against the governing criteria."

Appendix B. Offside Evaluator

The offside example originated from a seemingly simple question posed to a football expert: 'What is the rule for being offside in football?' The question was classified in the Remember x Factual category and appeared likely to produce a simple response. In practice, the resulting response contained numerous required and supporting claims, motivating the evaluator shown below.

Prompt: [What is the rule regarding offside in football?]
Category: [Remembering x Factual]
Benchmark: [IFAB Laws of the Game — Law 11: Offside]
Evaluator Version: [1.2]

Evaluate the response using the following criteria:

Required Claims

Begin required claims

[C1] **[Factual Assertion]** The response identifies IFAB Law 11 – Offside as the rule governing offside.

[C2] **[Definitional]** The response states that a player is in an offside position when, at the moment the ball is played or touched by a teammate, any part of the player's head, body, or feet is:

(a) in the opponents' half; and
(b) nearer to the opponents' goal line than both the ball and the second-last opponent.

Both conditions must be presented as jointly required, and the judgment must be anchored to the teammate's play or touch of the ball rather than to the moment the player receives it.

[C3] **[Factual Assertion]** The response states that the hands and arms are excluded when determining offside position.

[C4] **[Conceptual Relation]** The response states that being in an offside position is not, by itself, an offense.

[C5] **[Conceptual Relation]** The response explains that a player in an offside position is penalized only by becoming involved in active play through one or more of the following: interfering with play; interfering with an opponent; or gaining an advantage from the offside position.

[C6] **[Procedural Guard]** The response states that there is no offside offense when a player receives the ball directly from a goal kick, throw-in, or corner kick.

[C7] **[Procedural Guard]** The response states that a player in an offside position who receives the ball from an opponent who deliberately plays the ball is generally not penalized for the prior offside position, while preserving the exception for a deliberate save.

[C8] **[Factual Assertion]** The response states that an offside offense results in an indirect free kick for the defending team.

[C9] **[Contextual Assertion]** The response states that the indirect free kick is taken from the place where the offside offense occurred.

[C10] **[Cognitive Operation Conformance]** The response recalls and presents the governing offside rule rather than substituting an unrelated tactical explanation, opinion, or example for the rule itself.

End required claims.

Supporting Claims

These claims are optional. They are not required for score 1, but if included, they must be correct. **Begin supporting claims**

[S1] **[Contextual Assertion]** The response explains that interfering with play generally means playing or touching the ball.

[S2] **[Contextual Assertion]** The response explains that interfering with an opponent can include clearly obstructing an opponent's line of vision or challenging for the ball in a way that affects the opponent's ability to play the ball.

[S3] **[Conceptual Relation]** The response explains that gaining an advantage includes playing a ball that rebounds or is deliberately saved after the player was in an offside position.

[S4] **[Conceptual Relation]** The response explains that a player may commit an offside offense without touching the ball if the player's actions interfere with an opponent.

[S5] **[Definitional]** The response additionally notes that the halfway line is excluded from the opponents' half for purposes of offside position.

[S6] **[Contextual Assertion]** The response provides the shorthand that offside position plus involvement in active play equals an offside offense, provided it is presented as a summary rather than as a substitute for the more precise rule.

[S7] **[Conceptual Relation]** The response states that a player may legally remain in an offside position so long as the player does not become involved in active play in one of the ways specified by Law 11.

[S8] **[Citation Support]** The response cites or links to IFAB Law 11 or equivalent official IFAB material supporting the offside rule.

End supporting claims.

Disallowed Claims

The response must not include any of the following.
Begin disallowed claims

[D1] **[Factual Assertion — Disallowed]** The response states that merely being in an offside position automatically constitutes an offside offense.

[D2] **[Definitional — Disallowed]** The response states that a player is offside solely because the player is behind the defenders, without accounting for the ball, the second-last opponent, the opponents' half, or the relevant moment of judgment.

[D3] **[Definitional — Disallowed]** The response anchors the offside-position judgment to the moment the player receives the ball rather than the moment a teammate plays or touches it.

[D4] **[Definitional — Disallowed]** The response states or implies that a player on the halfway line is in an offside position, or that the halfway line counts as part of the opponents' half for offside purposes.

[D5] **[Definitional — Disallowed]** The response states or implies that a player level with the second-last opponent, or with the last two opponents, is in an offside position.

[D6] **[Factual Assertion — Disallowed]** The response includes the hands or arms when determining whether a player is in an offside position.

[D7] **[Conceptual Relation — Disallowed]** The response states that a player must touch the ball in order to commit any offside offense.

[D8] **[Conceptual Relation — Disallowed]** The response omits or contradicts the distinction between offside position and offside offense.

[D9] **[Procedural Guard — Disallowed]** The response states that receiving the ball directly from a goal kick, throw-in, or corner kick can itself create an offside offense.

[D10] **[Procedural Guard — Disallowed]** The response states without qualification that every touch by a defender resets offside; the distinction between a deliberate play and a deliberate save must be preserved.

[D11] **[Factual Assertion — Disallowed]** The response states that the normal restart for an offside offense is a direct free kick, penalty kick, goal kick, dropped ball, or other restart rather than an indirect free kick.

[D12] **[Citation Support — Disallowed]** The response cites a source as supporting an offside rule that the cited source does not actually establish.

End disallowed claims.

Scoring Rule

Return 1 if the response satisfies all required criteria, does not contradict them, and contains no incorrect, unsupported, or disallowed substantive claims. Supporting claims may be absent. If supporting claims are present, they must be correct and must correspond to one of the listed optional supporting criteria. Return 0 otherwise.

Appendix C. The Taxonomy Table for Law 11 Prompts

Knowledge Dimension	Factual	Conceptual	Procedural	Metacognitive
Remember	According to Law 11 of the IFAB Laws of the Game, what restart is awarded when a player commits an offside offense?	State the offside rule in football as defined by Law 11, covering offside position, offside offense, and the sanction.	List, in order, the steps an assistant referee follows to judge whether an offside offense has occurred.	Which offside judgments do the Laws of the Game explicitly leave to the discretion of the referee rather than defining objectively?
Understand	Explain what the term "second-last opponent" means in Law 11, including how the goalkeeper is counted.	Explain the difference between being in an offside position and committing an offside offense, including why one is penalized and the other is not.	Explain why offside position is fixed at the moment the ball is played or touched by a teammate, while the offense itself is assessed only later, when the player becomes involved in active play.	Explain why the Laws do not quantify how near an attacker must be to the goalkeeper in order to obstruct the goalkeeper's line of vision, and what this implies about the kind of judgment the referee must exercise.

Table C.3: The Complete Taxonomy Table with Prompts - Part 1

Knowledge Dimension	Factual	Conceptual	Procedural	Metacognitive
Apply	An attacker commits an offside offense two meters from the penalty spot, inside the opponents' penalty area. State the restart that is awarded and the location from which it is taken.	An attacker in an offside position never touches the ball but stands directly in the goalkeeper's line of vision as a teammate's shot enters the goal. Determine whether an offside offense has occurred and identify which mode of active involvement, if any, applies.	Team B's goalkeeper has advanced upfield to play the ball, so that exactly one Team B outfield defender is nearer to Team B's goal line than the goalkeeper. Applying the counting rule of Law 11, determine which Team B player now sets the offside line for Team A's attackers.	In a match, an attacker in an offside position stands roughly three meters from the goalkeeper, without moving, as a teammate's shot enters the goal. For this incident, state which elements of the offside decision the Laws determine objectively and which elements require referee judgment.
Analyze	A match report includes four rulings: offside interference, a careless trip, deliberate handball in the penalty area, and gaining an offside advantage after a rebound. Identify which rulings are governed by Law 11.	A goal is scored as follows: Attacker A1 is in an offside position when teammate A2 shoots; the shot rebounds off the goalpost to A1, who taps it in without any defender touching the ball. Analyze this goal to determine which mode of active involvement under Law 11 applies and whether the goal stands.	An assistant referee describes their procedure on a disallowed goal: "(1) I waited to see whether the attacker touched the ball. (2) When the attacker touched it, I judged the attacker's position at that moment of touch. (3) The attacker was behind the second-last opponent at that moment, so I raised my flag." Analyze this procedure and identify which step misapplies the offside procedure prescribed by Law 11.	Consider the following written analysis of an offside call: "The attacker was offside because he was level with the second-last defender, and level always means offside. Since he was within five meters of the goalkeeper, the Laws automatically deem this an obstruction of the line of vision." Distinguish the claims in this analysis that Law 11 actually establishes from the claims that follow only from the author's own assumptions.

Table C.4: The Complete Taxonomy Table with Prompts - Part 2

Knowledge Dimension	Factual	Conceptual	Procedural	Metacognitive
Evaluate	Evaluate the following claim and state whether it is correct: "An offside offense is punished with a direct free kick to the opposing team." Justify your verdict by reference to Law 11.	Evaluate whether the following interpretation of Law 11 is correct: "Any touch of the ball by a defender, deliberate or not, resets offside and puts a previously offside attacker onside." Support your evaluation with the distinction Law 11 draws between deliberate play and deflection.	A league's video-review protocol states: "First determine whether the attacker interfered with play or an opponent. Only if interference is confirmed, check whether the attacker was in an offside position at the moment of the teammate's touch." Evaluate whether this protocol respects the required relationship between the position assessment and the involvement assessment in Law 11.	A spectator was asked whether an attacker obstructed the goalkeeper's view and responded: "The attacker was two meters away, which the Laws define as obstruction, so this is definitively an offside offense." Evaluate whether this response hedged appropriately in the places where the Laws are silent, and identify any overclaim.
Create	Compile a fact sheet of every restart prescribed or referenced by Law 11, listing for each: the restart type, the team awarded the restart, and the location from which it is taken.	Create a taxonomy of offside scenarios organized by mode of active involvement (interfering with play, interfering with an opponent, gaining an advantage), with at least two distinct example scenarios per mode.	Create a decision framework, as an ordered checklist, that cleanly partitions the offside position test from the offside offense test, specifying for each test the exact moment in play at which it is assessed.	Create guidance for referees describing where Law 11 requires subjective judgment.

Table C.5: The Complete Taxonomy Table with Prompts - Part 3